

Test of Mathematics for University Admission
Mock Set H Paper 1
 Applications of Mathematical Knowledge

Time allowed: **75 minutes**
 Questions: 20 multiple choice
 Calculators: not permitted
 Written by: Kevin Hu
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Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Every question uses AS content only. The difficulty lies in how that content is used: parameters that change the case, conditions that quietly remove a solution, and answers that come from structure rather than from brute force.

[illegible]

Question 1

Find the minimum value, over all real x , of

$$|x - 1| + |2x - 3| + |3x - 6|.$$

- A $\frac{1}{2}$
- B 1
- C $\frac{3}{2}$
- D 2
- E $\frac{5}{2}$
- F 3

Question 2

A sequence is defined by $a_1 = 1$ and

$$a_{n+1} = \frac{a_n}{1 + n a_n} \quad (n \geq 1).$$

Find $\frac{1}{a_{100}}$.

- A 4950
- B 4951
- C 5050
- D 5051
- E 9901
- F 100

Question 3

How many ordered pairs of integers (x, y) satisfy

$$x^2 + xy + y^2 = 7?$$

- A 4
- B 6
- C 8
- D 10
- E 12
- F 16

Question 4

The real number x satisfies $9^x - 6^x = 4^x$. Find the value of

$$\left(\frac{3}{2}\right)^x - \left(\frac{2}{3}\right)^x.$$

- A 1
- B $\sqrt{5}$
- C $\frac{1 + \sqrt{5}}{2}$
- D $\frac{\sqrt{5} - 1}{2}$
- E 2
- F $\frac{3}{2}$

Question 5

A triangle has sides of length 4, 5 and 6. Find the ratio of its largest angle to its smallest angle.

- A $\frac{3}{2}$
- B $\frac{5}{3}$
- C 2
- D $\frac{5}{2}$
- E 3
- F the ratio is irrational

Question 6

Two circles, of radii 3 and 5, intersect at the points P and Q . At P the tangents to the two circles are perpendicular to each other. Find the length of the common chord PQ .

- A $\frac{15}{4}$
- B $\frac{15\sqrt{34}}{17}$
- C $\sqrt{34}$
- D $\frac{30}{\sqrt{17}}$
- E 4
- F $\frac{15\sqrt{2}}{4}$

Question 7

A circle passes through the three points of the parabola $y = x^2$ whose x -coordinates are 1, 2 and -4 . Which of the following is true?

- A The circle meets the parabola at a fourth point, with x -coordinate -1 .
- B The circle meets the parabola at a fourth point, with x -coordinate 3.
- C The circle meets the parabola at a fourth point, with x -coordinate $\frac{1}{2}$.
- D The circle meets the parabola at two further points.
- E No circle passes through all three points.
- F The circle meets the parabola at no further point: it is tangent to the parabola at $(1, 1)$.

Question 8

Find the area of the region of the plane enclosed by the curve

$$|x - 2| + |y - 3| + |x + y - 5| = 4.$$

- A 8
- B $4\sqrt{3}$
- C 10
- D 12
- E 16
- F 18

Question 9

In triangle ABC , $BC = 5$, $CA = 6$ and $AB = 7$. The bisector of angle C meets AB at D . Find the length CD .

- A $\frac{12\sqrt{15}}{11}$
- B $\frac{6\sqrt{10}}{5}$
- C $\frac{2\sqrt{30}}{3}$
- D $\sqrt{15}$
- E $\frac{35}{11}$
- F $\frac{12\sqrt{5}}{5}$

Question 10

Evaluate

$$\sum_{k=1}^{99} \frac{1}{\sqrt{k} \sqrt{k+1} (\sqrt{k} + \sqrt{k+1})}.$$

- A $\frac{99}{100}$
- B $\frac{1}{10}$
- C $\frac{9}{10}$
- D $\sqrt{99} - 1$
- E 1
- F $\frac{9}{\sqrt{10}}$

Question 11

Let r_1, r_2, r_3, r_4 be the four roots of $x^4 - 3x^2 + x + 1 = 0$. Find the value of

$$r_1^5 + r_2^5 + r_3^5 + r_4^5.$$

- A -9
- B 9
- C 14
- D -14
- E -15
- F 15

Question 12

From the point $(1, -3)$ two tangents are drawn to the parabola $y = x^2$. Find the area of the finite region bounded by the parabola and these two tangents.

- A $\frac{8}{3}$
- B $\frac{16}{3}$
- C $\frac{32}{3}$
- D 4
- E 8
- F $\frac{64}{3}$

Question 13

Evaluate

$$\sum_{k=1}^{10} \frac{k}{k^4 + k^2 + 1}.$$

- A $\frac{55}{111}$
- B $\frac{110}{111}$
- C $\frac{1}{2}$
- D $\frac{100}{201}$
- E $\frac{50}{101}$
- F $\frac{55}{112}$

Question 14

Find the minimum value, over all real x and y , of

$$\sqrt{x^2 + 1} + \sqrt{(y - x)^2 + 4} + \sqrt{(6 - y)^2 + 9}.$$

- A 6
- B $3 + \sqrt{13}$
- C $6\sqrt{2}$
- D 9
- E $2\sqrt{17}$
- F $\sqrt{37} + \sqrt{5}$

Question 15

Find the set of all real numbers x for which x , x^2 and x^3 are the side lengths of a non-degenerate triangle.

- A $0 < x < \frac{1 + \sqrt{5}}{2}$
- B $\frac{1}{2} < x < 2$
- C $1 < x < \frac{1 + \sqrt{5}}{2}$
- D $\frac{\sqrt{5} - 1}{2} < x < \frac{1 + \sqrt{5}}{2}$
- E $\frac{\sqrt{5} - 1}{2} < x < 1$
- F $x = 1$ only

Question 16

Evaluate the infinite series

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = \frac{1}{2} + \frac{4}{4} + \frac{9}{8} + \frac{16}{16} + \cdots.$$

- A 2
- B 3
- C 4
- D $\frac{9}{2}$
- E 5
- F 6

Question 17

Find the total area of the finite regions enclosed between the curves

$$y = |x^2 - 4| \quad \text{and} \quad y = |x| + 2.$$

- A $\frac{13}{3}$
- B $\frac{19}{3}$
- C 9
- D $\frac{29}{3}$
- E $\frac{31}{3}$
- F $\frac{37}{3}$

Question 18

The tangent to the curve $y = x^3$ at a point P (not the origin) meets the curve again at Q . The tangent at Q meets the curve again at R . Find the ratio of the area of the region enclosed between the curve and the segment PQ to the area of the region enclosed between the curve and the segment QR .

- A $\frac{1}{8}$
- B $\frac{1}{16}$
- C $\frac{1}{4}$
- D $\frac{1}{32}$
- E $\frac{1}{27}$
- F it depends on P

Question 19

For which values of k does the equation

$$|x^2 - 4x| = x + k$$

have exactly four real solutions?

- A $0 \leq k \leq \frac{9}{4}$
- B $-4 < k < 0$
- C $0 < k < \frac{9}{4}$
- D $0 < k < 4$
- E $k > \frac{9}{4}$
- F $-4 < k < \frac{9}{4}$

Question 20

For each positive integer n , let $m(n)$ be the minimum value, over all real x , of

$$f_n(x) = |x - 1| + |x - 2| + |x - 3| + \cdots + |x - n|.$$

For how many integers n with $1 \leq n \leq 50$ is $m(n)$ a perfect square (including 0)?

- A** 7
- B** 24
- C** 25
- D** 26
- E** 36
- F** 50

END OF PAPER